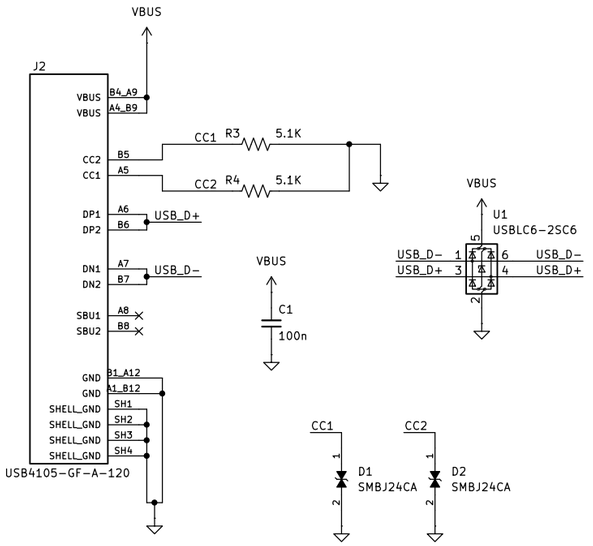
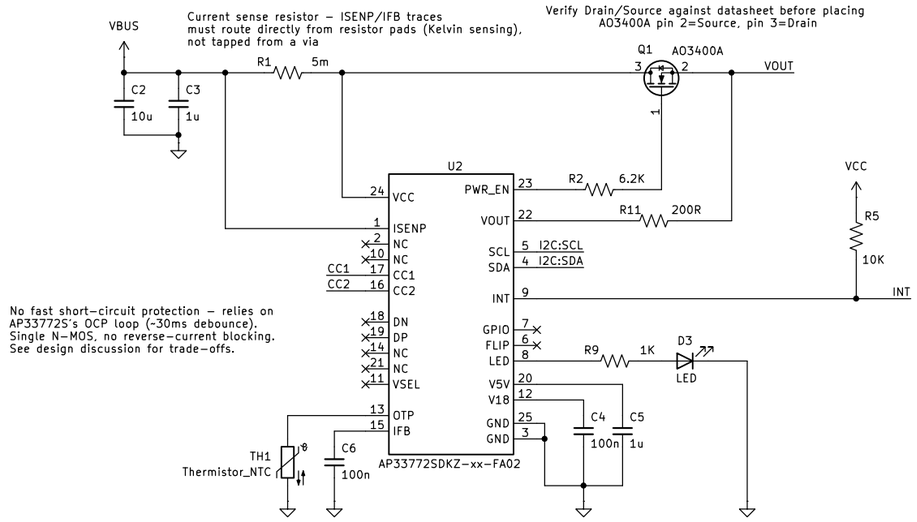
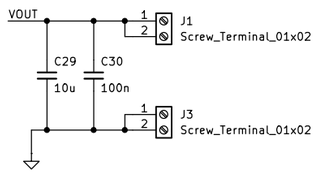


USB-C INPUT FOR POWER AND DATA



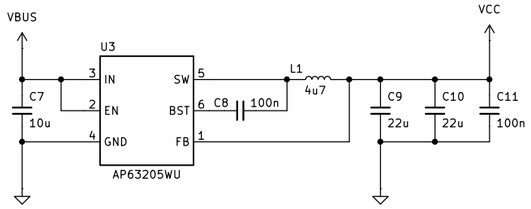
COMPONENT NAME	COMPONENT FUNCTION
SW1	DECREASE VOLTAGE
SW2	DECREASE CURRENT
SW3	OUTPUT ON/OFF
SW4	INCREASE VOLTAGE
SW5	INCREASE CURRENT
SW6	MCU RESET
J1	VOUT +
J2	USB-C CONN
J3	VOUT –
J4	I2C OLED CONN

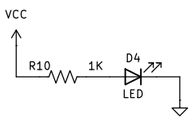
State	LED Indication	VOUT	Comments
INIT	NA	OFF	VBUS/Rp attached and AP33772S initialization
CHARGING	4-sec Breathing	ON	Successful negotiation or enter Non-PD Mode and start charging
MISMATCH	Full Light	OFF	VSELMIN mismatch (VREQ < VSELMIN)
MOISTURE	2-sec Flicker	OFF	Abnormal impedance detected in the Type-C connector
FAULT	0.6-sec Flicker	OFF	OVP, OCP, UVP or OTP occurs

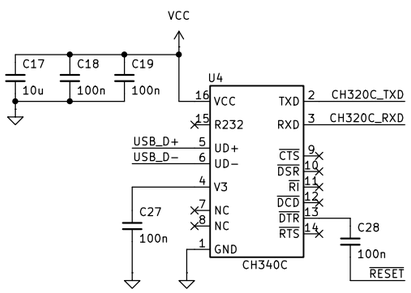


Firmware-enforced minimum output: 5V
(headroom for buck regulator dropout – do not allow lower in firmware)

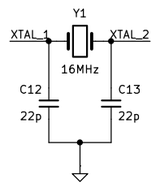
AP33772S PROGRAMMABLE POWER SUPPLY AND TERMINALS

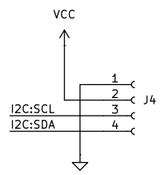


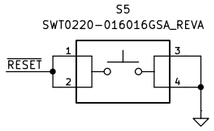


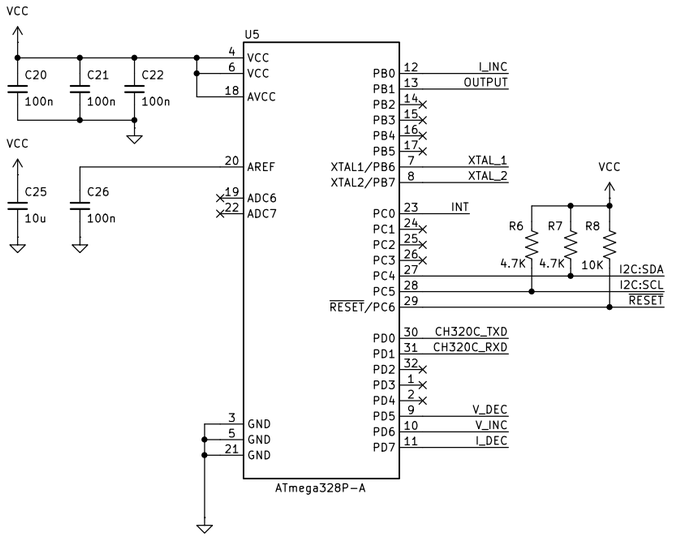


PD0/PD1 shared with CH340C – no separate serial debug pin available

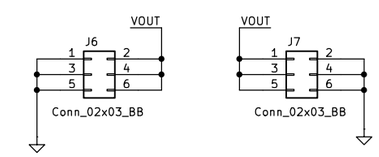




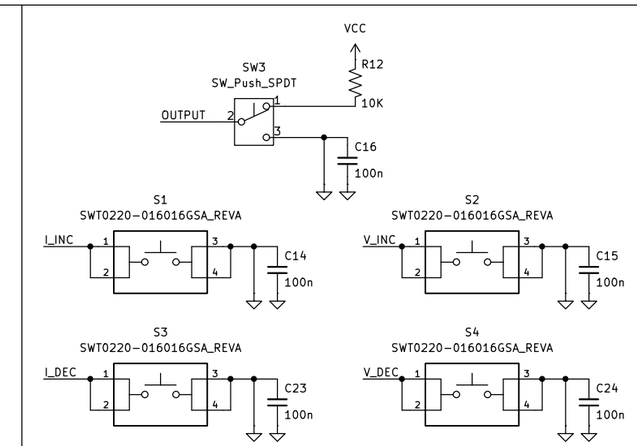




ATMEGA328P-A AND RELATED CIRCUITRY



PCB MOUNT HEADERS



CONTROL CIRCUITRY

DESIGN NOTES

- PPS trigger board, not a regulated bench supply – output tracks the charger's PPS contract, no local CV/CC control loop
- Programming: ICSP only for bootloader; CH340C/UART for day-to-day flashing after that
- No crystal-free option – Y1 required for reliable UART baud rate
- I2C bus shared: AP33772S + OLED, single 4.7k pull-up pair
- Output: dual 2-pin screw terminals, both wired to the same VOUT/GND net, for multi-load use

NANO PPS – ATMEGA328P-A CONTROLLED AP33772S PPS TO DELIVER POWER UPTO 45 WATTS.

NANO PPS